STAFF REVIEW · H446 318 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Current instruction.

2. Reference to a key in another table.

3. Resolving a domain name to an IP address.

4. $(A \vee \neg A) \wedge (A \vee B) = A \vee B$.

I Do — source worked steps

1. Explain the method — MAR holds address, MDR transfers contents, and CIR holds instruction for decoding.

We Do — guided CHECK

1. Explain the method
Use exactly eight bits; inversion alone represents the wrong number.

You Do — source mark schemes

1. MAR=70 [1]; the instruction LOAD 25 is fetched from M[70] into MDR [2]; CIR receives LOAD 25 from MDR [1]; PC advances to 71 [1]; the control unit decodes the instruction in CIR [1].

2. +18=00010010 [2], invert11101101 [2], add1=11101110 [1], leading1 indicates negative [1].

3. SELECT Student.Name,Mark.Score [1] FROM Student [1] INNER JOIN Mark [1] ON Student.ID=Mark.ID [2]; output Ben80 [1].

Exit Ticket — source answer

1. False. Explain the method on new values.